

Nowcasting of GDP: International Experience, Methodology, and Evaluation

Master Thesis - Part I

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Chapter 1

Research Overview and Theoretical Foundations

1.1 Background, Motivation and Research Gap

Among the central challenges faced by modern macroeconomic analysis is the persistent lag with which official statistics measure the state of real economic activity. Gross Domestic Product (GDP)—the most widely used indicator of aggregate output—is reported quarterly and typically published with a delay of four to eight weeks following the end of the reference period. During this window, monetary authorities, fiscal planners, and financial market participants must form judgements about the current economic conjuncture on the basis of data that may already be considerably outdated. This information deficit takes on particular significance during episodes of rapid change, when waiting for official releases can delay essential policy responses by an entire quarter or more.

The concept of *nowcasting*—predicting the present, the very near future, and the very recent past of a target variable—has emerged as a systematic response to this challenge. Originally a term from meteorology, it was formally imported into economics by Giannone, Reichlin, and Small, who demonstrated how the continuous flow of monthly and higher-frequency statistical releases could be exploited to produce timely estimates of quarterly GDP before the official figure becomes available.[?] Unlike conventional medium-term forecasting, nowcasting is explicitly oriented toward real-time data assimilation: each new

release of industrial production, survey indices, or financial variables is treated as a signal that revises the current assessment of aggregate output.

Although the nowcasting literature is mature and well-developed for advanced economies—particularly the United States and the euro area—a less-examined question concerns the relative performance of traditional econometric frameworks versus modern machine learning (ML) approaches, especially in emerging and transition economies where data are scarcer, structural breaks are more frequent, and institutional capacity differs substantially. The informational value of unstructured digital data, such as internet search volumes, also remains inadequately explored in economies where available indicators and internet connectivity differ from those of high-income countries. This thesis addresses these gaps through a comparative synthesis drawing on evidence from six economies: the Netherlands, the United States, Peru, Georgia, Armenia, and Belarus.

1.2 Research Questions, Contributions and Analytical Scope

Three core research questions organise the enquiry. First, how do classical econometric models—bridge equations, mixed-data sampling (MIDAS) regressions, and Dynamic Factor Models (DFMs)—compare to machine learning algorithms in terms of point nowcast accuracy for GDP at very short horizons? Second, do non-linear and high-dimensional models such as random forests, support vector machines, and recurrent neural networks confer particular advantages during episodes of structural volatility, including the global financial crisis and the COVID-19 pandemic? Third, can forecast combination strategies and model interpretability tools—most notably Shapley value decompositions—extend the practical utility of nowcasts beyond what any single model delivers?[?]

The thesis makes contributions at three levels. **Methodologically**, it provides a systematic head-to-head comparison of traditional and ML-based nowcasting specifications applied within a unified pseudo-real-time evaluation framework. **Empirically**, it documents model performance across distinct phases of the business cycle, distinguishing the behaviour of competing approaches in tranquil periods from their behaviour during crisis episodes. **Practically**, it assesses whether ensemble combinations and transparency tools can render complex ML nowcasts sufficiently interpretable for day-to-day macroeconomic surveillance.

The analytical scope is restricted to short-horizon nowcasting of real GDP growth for a single country case study, with the sample period determined by the joint availability of quarterly national accounts and a sufficiently rich set of monthly indicators. Three simplifying assumptions apply throughout: a *pseudo-real-time* design that replicates real-time information flow using final-vintage data combined with historical release calendars;

approximate stationarity achieved through standard transformations (logarithms, differencing, seasonal adjustment); and a finite evaluation window that may encompass only a limited number of crisis observations, constraining the statistical power of state-dependent comparisons.

Chapter 2

International Experience of Nowcasting

2.1 Nowcasting: Definition, Data and Modelling Strategies

At its core, the nowcasting problem arises from a fundamental temporal mismatch: the variable of primary interest—GDP—is released with a lag of one to two months following the close of the quarter, whereas a large and diverse set of related indicators becomes available on a monthly, weekly, or even daily basis throughout that same quarter. Giannone et al. formalise this as a problem of optimal filtering: the nowcast of y_t is the best statistical estimate given the information set Ω_t available at calendar time t , where Ω_t contains different series observed up to different points in time—producing a dataset with a characteristic “ragged edge” at its right-hand boundary.[?] The nowcaster’s task is therefore to process an asynchronous and growing information set in a way that is both statistically coherent and computationally feasible in real time.

A key analytical concept emerging from this framework is the notion of *news*: when a new data release arrives, its contribution to revising the nowcast can be decomposed into two components. *Timeliness* captures the informational value derived simply from arriving earlier than other indicators. *Economic content* captures the intrinsic explanatory power of the series for GDP once the timing advantage is controlled for. Survey indicators—such as the Federal Reserve Bank of Philadelphia’s business outlook survey—carry a disproportionately large impact on real-time nowcast revisions compared with the Employment Report, precisely because they arrive earlier in the publication cycle. Once timeliness is controlled for, however, hard data like industrial production and retail sales prove highly significant

in terms of economic content. This distinction has important practical implications: the optimal response to a new data release is not uniform but depends on its location in the publication calendar.

Four broad families of indicators populate a typical nowcasting dataset. **Hard macroeconomic data**—industrial production, retail and wholesale trade, employment, exports and imports—are direct components or close proxies of the national accounts, making them the most informative signals for GDP once released. **Soft survey-based indicators**—purchasing managers’ indices, business tendency surveys, consumer confidence—react more quickly to news but may embed subjective noise. **Financial variables**—yield spreads, equity indices, credit volumes, exchange rates—are available continuously and encode forward-looking market assessments. **Alternative and high-frequency data**—Google Trends search volumes, payment card records, mobility indices—are especially relevant at the start of a quarter when conventional statistics are absent.[? ?]

Three modelling strategies dominate the literature for handling mixed-frequency data. *Bridge equations* aggregate monthly predictors to quarterly frequency and estimate a direct projection of GDP, providing a transparent but timing-limited approach. *MIDAS* regressions retain the original high-frequency observations through parsimonious distributed-lag polynomial weights. Most powerfully, *Dynamic Factor Models* in state-space form use the Kalman filter to handle the ragged edge directly, simultaneously extracting latent common factors from a large panel and updating nowcast estimates as each new data release arrives.[?] Machine learning methods address the ragged edge by constructing balanced design matrices through lagging or last-observation-carried-forward imputation, making them applicable without modifying the core estimation routine.

2.2 International Experience: Advanced and Emerging Economies

In advanced economies, central banks were among the first to institutionalise systematic nowcasting. The Federal Reserve System and the European Central Bank both deploy variants of the factor-model framework, using panels of two hundred or more monthly series to produce continuous updates of current-quarter GDP. Bok et al. provide a comprehensive account of how such a system is operationalised in practice at the Federal Reserve Bank of New York, maintaining a tracking database of approximately 38,000 monthly and quarterly economic series, automating the data ingestion pipeline, and formally quantifying the “news” contribution of each scheduled release to the overall nowcast revision.[?] Their framework highlights that the number of relevant latent factors need not be fixed in advance but can be selected through information criteria applied to the full data panel, a practical refinement over earlier implementations. A complementary dimension of real-time

analysis is documented by Anesti, Galvão, and Miranda-Agrippino, who show that the quality of preliminary nowcasts depends partly on how substantially initial GDP releases are later revised—implying that model rankings evaluated on final-vintage data may look different from rankings obtained in genuine real time.[?]

For the Netherlands, Kant, Pick and de Winter compare DFMs against a suite of machine learning methods over 1992–2018, finding that the Random Forest delivers the most accurate nowcasts for the current quarter—an RMSFE of 0.91 in month 1 versus 0.98 for the DFM relative to the prevailing mean benchmark—while the DFM recovers its advantage for backcasting once hard data on the target quarter begin to arrive (RMSFE 0.79 versus 0.88).[?] Studies for Armenia and Belarus demonstrate that LASSO (Armenia: RMSE 3.38) and SVM (Belarus: RMSE 1.44) minimise forecast errors for real GDP, because growth in both countries is governed by a narrow set of dominant variables whose co-movement with GDP is captured efficiently by sparse linear specifications; for the GDP deflator, Recurrent Neural Networks achieve RMSE values of 2.16 and 1.05 respectively, because inflation exhibits strong path-dependence and non-linear momentum that RNNs capture through accumulated temporal context.[?] In Peru, Gradient Boosting Machines incorporating Google Trends proxies achieve an RMSE of 0.17—compared with 2.55 for the AR(1) baseline—and successfully anticipated the output contraction of March 2020 weeks before official statistics confirmed it.[?] In Georgia, ML algorithms outperform the traditional DFM by 17–21 percent in RMSE, with gains concentrated in periods incorporating VAT turnover and Google Trends indices as high-frequency “Type-1” data available within the first week of the following month.[?] In the United States, a Bayesian Structural Time Series model augmented with 155 Google Trends search-volume categories reduces RMSFE by up to 40 percent in the first two months of the quarter—the period

when conventional hard indicators are almost entirely unavailable.[?]

Chapter 3

Why Nowcasting is Important

3.1 Policy, Market and Crisis Relevance

The operational relevance of nowcasting derives principally from the asymmetry between the pace at which economic conditions evolve and the pace at which official measurement systems document those conditions. For monetary authorities, real-time estimates of GDP growth and inflation are essential inputs to interest-rate decisions and forward guidance communications: because policy acts on the economy with a lag of several quarters, the assessment of the *current* conjuncture—including the economy’s position relative to potential output—is at least as important as the medium-term forecast. Nowcasting systems allow central banks to quantify how each individual data release shifts the estimate of current-quarter GDP—the *news* released by each new data point—enabling a systematic and reproducible real-time process in place of informal data watching.[?]

A particularly important but underappreciated dimension of real-time analysis is that policymakers face a *double uncertainty*: they must act not only on the basis of delayed data, but on the basis of data that will itself be substantially revised in subsequent releases. Anesti et al. document this for the United Kingdom, demonstrating that the gap between preliminary and final-vintage GDP estimates is large enough to affect the sign of quarterly growth in a non-trivial share of observations, which in turn implies that real-time nowcast rankings across models may look quite different from rankings obtained with revised data.[?] This finding underscores why evaluating models in a genuinely pseudo-real-time setting matters greatly for drawing policy-relevant conclusions. Fiscal authorities face a parallel need: revenue projections, budget balance estimates, and debt-sustainability analyses all depend on the current level of economic activity, and erroneous real-time estimates can lead to premature tightening or unnecessarily delayed stimulus.

During the COVID-19 pandemic, nowcasting frameworks enabled governments in the Netherlands, Georgia, and Armenia to calibrate emergency fiscal packages to an estimate of the output decline that preceded official GDP data by an entire quarter. Crucially,

machine learning models proved most valuable precisely during the most severe episodes: the Netherlands study finds that the Random Forest’s advantage over the DFM was most pronounced both during the COVID-19 period and during the global financial crisis of 2008–2009—precisely the episodes where the non-linear, regime-switching behaviour of the economy fell outside the range of experience that linear factor models had been trained to capture.[?] Financial market participants constitute a second major constituency: the BSTS model of Kohns and Bhattacharjee shows that incorporating Google Trends data produces materially sharper estimates of US GDP in the weeks before any hard data are available, allowing market participants to update yield and equity price expectations with better information.[?]

3.2 Limitations and Critical Perspectives

The practical value of nowcasting models must be assessed alongside a clear-eyed awareness of their structural limitations. A fundamental constraint is data quality: measurement error, subsequent revisions, definitional reclassifications, and unstable seasonal adjustment procedures can corrupt the signal extracted from raw indicators, with these problems being more acute in emerging economies where data infrastructure is still developing. The estimated relationships between high-frequency predictors and quarterly GDP are reduced-form in nature and therefore susceptible to the Lucas critique: a change in policy regime or in agents’ behaviour can alter the co-movement structure that the model was trained to exploit.[?]

Alternative data sources introduce specific risks of their own. Internet search volumes may be distorted by algorithmic changes to search platforms, shifts in the demographic composition of internet users, or the emergence of competing information sources—any of which can create structural breaks in the relationship between search indices and the underlying economic phenomena they proxy. Highly flexible machine learning architectures furthermore risk becoming opaque “black boxes” that generate accurate in-sample estimates but lack interpretability, making them difficult to communicate credibly to policymakers and potentially prone to overfitting when applied to economic regimes not represented in the training data. These concerns strengthen the case for rigorous out-of-sample evaluation, transparent model documentation, ensemble approaches that diversify across model families, and the sustained role of informed economic judgement alongside statistical

output.[?]

Chapter 4

Analytical and Data Framework

4.1 Econometric and Machine Learning Methodology

The Dynamic Factor Model occupies the foundational position in central-bank nowcasting practice because it simultaneously addresses the curse of dimensionality and the ragged-edge problem. In its canonical formulation, the co-movement of a large $N \times 1$ vector of standardised macroeconomic series X_t is driven by a small $r \times 1$ vector of latent common factors F_t through the **observation equation**

$$X_t = \Lambda F_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \Psi), \quad (4.1)$$

where Λ is the $N \times r$ matrix of factor loadings and ε_t is the vector of idiosyncratic disturbances. The factor dynamics follow the **transition equation**

$$F_t = AF_{t-1} + u_t, \quad u_t \sim \mathcal{N}(0, Q), \quad (4.2)$$

and the full system is estimated by the Kalman filter, which provides the minimum mean-squared-error estimate of F_t conditional on observed data, handling missing observations at the ragged edge by skipping the measurement update for unreleased series.[?] From an economic standpoint, the latent factors can frequently be mapped to interpretable macro-financial concepts—a global real activity factor, a financial conditions factor, a price-pressure factor—each capturing a dimension of the business cycle as documented in the large-scale operational implementation at the Federal Reserve Bank of New York.[?]

Machine learning approaches extend the nowcasting toolkit by relaxing linearity and sparsity restrictions. *Penalised regressions*—LASSO ($\hat{\beta} = \arg \min_{\beta} \{\|y - X\beta\|^2 + \lambda\|\beta\|_1\}$) and Ridge—shrink or zero out coefficients of less informative predictors, performing automatic variable selection or multicollinearity handling respectively. *Random Forests* grow an ensemble of B regression trees on bootstrapped samples and random predictor subsets, averaging their predictions $\hat{f}_{RF}(x) = B^{-1} \sum_{b=1}^B \hat{f}_b(x)$ to capture non-linear regime

dependencies and interaction effects.[?]] *Recurrent Neural Networks* maintain a hidden state $h_t = \sigma(W_{hh}h_{t-1} + W_{hx}x_t)$ that accumulates temporal information across time steps, making them well-suited to persistent and non-linear price dynamics. When individual model predictions are further combined with inverse-RMSE weights, the resulting ensemble typically outperforms any single specification—a pattern confirmed across the Armenia/Belarus and Georgian country studies.[?]]

4.2 Data Infrastructure, Transformations and Feature Engineering

A nowcasting dataset integrates indicators spanning multiple frequencies. *Real-activity hard data*—industrial production, retail trade indices, exports, imports, employment—are the most directly relevant because they measure components or close analogues of the national expenditure accounts, but are subject to publication lags of three to six weeks. *Soft survey indicators*—PMIs, business tendency and consumer confidence surveys—are released faster but embed subjective reporting bias. *Financial market variables*—term spreads, equity indices, credit aggregates, exchange rates—are available in real time and encode forward-looking expectations. *Alternative high-frequency data*—Google Trends indices, card-transaction aggregates, mobility records—are particularly valuable at the very start of a quarter before any conventional monthly statistics are published.[?]]

Preparing these heterogeneous series for modelling requires a structured sequence of transformations. Level variables are converted to log-differences or growth rates to achieve approximate stationarity; seasonal and trading-day effects are removed using standard filter-based decompositions; all series are then standardised (zero mean, unit variance) before entry into factor models or penalised regressions. Pseudo-real-time evaluation demands an additional layer of discipline: a release calendar for each series must be reconstructed from historical records so that each nowcast uses only data that would genuinely have been available at the corresponding point in time.[?]] Feature engineering further enriches the predictor set: nominal trade and retail series are deflated to real quantities; real interest rates are computed by subtracting expected inflation; and Google Trends indices are grouped into interpretable economic categories—labour-market stress, consumption activity, financial distress—and de-seasonalised before entry into the model.[?]

Chapter 5

Cross-Country Evidence and Comparative Assessment

5.1 Evidence from Developing Economies: Peru and Georgia

Tenorio and Perez examine GDP nowcasting for Peru using a dataset spanning January 2007 to May 2023, combining 91 structured macroeconomic indicators with 10 groups of Google Trends variables carefully designed to reflect distinct economic mechanisms: consumption behaviour is proxied by brand-level queries (searches for “Kia”, “Toyota”, “Restaurants”, “Cinema”), financial conditions by loan-market searches (“Credits”, “Loans”, “Mortgages”, “Deals”), and external demand by travel-related and climate-event queries (“flights”, “visa”, “El Niño”).[?] Across a pseudo-real-time evaluation, Gradient Boosting Machines achieve an RMSE of 0.17 compared with 0.32 for the best-performing DFM variant and 2.55 for the AR(1) baseline—a reduction in forecast error of more than 20–25 percent relative to conventional benchmarks. The decisive advantage of the ML approach is most visible in two situations: first, in the weeks immediately following a quarter’s opening when hard data remain entirely unavailable and only search-index signals carry information; and second, during the COVID-19 shock of early 2020, when models incorporating unstructured sentiment data correctly anticipated the magnitude of the contraction weeks before official statistics confirmed it. A key auxiliary finding is that combining structured and unstructured predictors delivers gains exceeding what either data type achieves in isolation, confirming a complementarity between conventional macroeconomic indicators and digital footprint data.

Mgebrishvili’s investigation of Georgia draws on a granular monthly dataset that distinguishes “Type-1” indicators—exchange rates, interbank rates, and tax revenues available within one week of month-end—from “Type-2” indicators, including VAT turnover of enterprises, tourism receipts, and remittances published around mid-month; Google

Trends search category indices are incorporated alongside both types of conventional indicators as a separate alternative data source.[?] Among the ensemble of ML methods evaluated—LASSO, Ridge, Support Vector Regression, Linear SVM, Neural Networks, XGBoost, CatBoost, AdaBoost, and Stacked Generalisation—the ML ensemble collectively outperforms the DFM by 17–21 percent in RMSE on average, with the gap widening substantially during the 2020–2021 pandemic episode. Whereas DFMs correctly identify the sign of quarterly GDP growth in roughly two-thirds of quarters under normal conditions, the best ML specifications achieve directional accuracy above 70 percent even during turning-point episodes. Stacked Generalisation—which combines the predictions of multiple base learners through a meta-model trained on held-out data—proved especially robust at turning points, suggesting that ensemble aggregation across heterogeneous model families captures complementary aspects of the economic signal.

5.2 Evidence from Transition and Advanced Economies: Armenia, Belarus, Netherlands, and United States

Tsukarev, Poghosyan, Lemba and Grishin conduct an exhaustive model horse race for Armenia and Belarus, pitting bridge equations, MIDAS, and DFM against Ridge, LASSO, Elastic Net, Bagging, Random Forest (XGBoost), SVM, MLP, LSTM, and RNN.[?] One of the study’s headline findings is that machine learning algorithms outperform conventional econometric models in every single tested specification for both countries—a 100 percent win rate for ML across all target variables and forecast horizons. For real GDP nowcasting, LASSO attains the lowest RMSE in Armenia (3.38) and SVM dominates in Belarus (1.44): growth cycles in both countries are governed by a narrow set of drivers—commodity revenues, remittances, and sectoral production—whose co-movement with GDP is captured efficiently by sparse linear specifications that select these drivers while discarding the many near-collinear indicators that add only noise. For the GDP deflator, RNNs achieve RMSE values of 2.16 (Armenia) and 1.05 (Belarus), outperforming all competing methods, because inflation exhibits highly path-dependent dynamics with non-linear momentum that RNNs’ recurrent hidden state captures through accumulated temporal context. These contrasting results—sparse linear models for real output, recurrent non-linear models for prices—illustrate a broader principle: the optimal model architecture is determined not just by the country context but by the specific statistical properties of the target series. A combined nowcast averaging predictions across top-performing ML models with inverse-RMSE weights provides the most stable out-of-sample performance overall, confirming that diversification across model families is a reliable hedge against individual specification failures.

Kant, Pick and de Winter evaluate eleven methods for the Netherlands over 1992Q1–2018Q4.[?] At month-1 of the nowcast horizon, Random Forest produces an RMSFE of 0.91 versus 0.98 for DFM. The non-linearities captured by Random Forest are most valuable when data are sparse: soft indicators (confidence surveys, PMIs) dominate the predictor set early in the quarter and their relationship with GDP changes non-linearly across business-cycle phases. Once the quarter closes and hard data flow in, the DFM recovers its advantage, achieving an RMSFE of 0.79 at the backcast horizon versus 0.88 for Random Forest. Shapley value analysis reveals that variable importance shifts systematically: survey-based soft indicators dominate SHAP contributions during nowcasting, while industrial production and trade statistics account for the majority of predictive weight at backcasting horizons. For the United States, Kohns and Bhattacharjee’s BSTS model augmented with 155 Google Trends topic-level search indices—selected via the Horseshoe shrinkage prior—reduces RMSFE by up to 40 percent in the first two months of the quarter, with economy-related news, investing behaviour, and labour-market proxy categories carrying the highest posterior inclusion probabilities, confirming that internet-search behaviour provides an almost-instantaneous proxy for GDP sentiment that hard statistics will only document weeks later.[?]

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